

2 Functions on the complex plane

2.1 Continuous functions

Let f be a function defined on a set Ω of complex numbers. We say that f is **continuous** at the point $z_0 \in \Omega$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $z \in \Omega$ and $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \epsilon$. An equivalent definition is that for every sequence $\{z_1, z_2, \dots\} \subset \Omega$ such that $\lim z_n = z_0$, then $\lim f(z_n) = f(z_0)$.

The function f is said to be continuous on Ω if it is continuous at every point of Ω . Sums and products of continuous functions are also continuous.

Since the notions of convergence for complex numbers and points in $\mathbb{R}^2$ are the same, the function f of the complex argument $z = x + iy$ is continuous if and only if it is continuous viewed as a function of the two real variables x and y .

By the triangle inequality, it is immediate that if f is continuous, then the real-valued function defined by $z \mapsto |f(z)|$ is continuous. We say that f attains a **maximum** at the point $z_0 \in \Omega$ if

$$|f(z)| \leq |f(z_0)| \quad \text{for all } z \in \Omega,$$

with the inequality reversed for the definition of a **minimum**.

Theorem 2.1 *A continuous function on a compact set Ω is bounded and attains a maximum and minimum on Ω .*

This is of course analogous to the situation of functions of a real variable, and we shall not repeat the simple proof here.

2.2 Holomorphic functions

We now present a notion that is central to complex analysis, and in distinction to our previous discussion we introduce a definition that is genuinely *complex* in nature.

Let Ω be an open set in $\mathbb{C}$ and f a complex-valued function on Ω . The function f is **holomorphic at the point** $z_0 \in \Omega$ if the quotient

$$(1) \quad \frac{f(z_0 + h) - f(z_0)}{h}$$

converges to a limit when $h \rightarrow 0$. Here $h \in \mathbb{C}$ and $h \neq 0$ with $z_0 + h \in \Omega$, so that the quotient is well defined. The limit of the quotient, when it exists, is denoted by $f'(z_0)$, and is called the **derivative of f at z_0** :

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}.$$

It should be emphasized that in the above limit, h is a complex number that may approach 0 from any direction.

The function f is said to be **holomorphic on Ω** if f is holomorphic at every point of Ω . If C is a closed subset of $\mathbb{C}$, we say that f is **holomorphic on C** if f is holomorphic in some open set containing C . Finally, if f is holomorphic in all of $\mathbb{C}$ we say that f is **entire**.

Sometimes the terms **regular** or **complex differentiable** are used instead of holomorphic. The latter is natural in view of (1) which mimics the usual definition of the derivative of a function of one real variable. But despite this resemblance, a holomorphic function of one complex variable will satisfy much stronger properties than a differentiable function of one real variable. For example, a holomorphic function will actually be infinitely many times complex differentiable, that is, the existence of the first derivative will guarantee the existence of derivatives of any order. This is in contrast with functions of one real variable, since there are differentiable functions that do not have two derivatives. In fact more is true: every holomorphic function is analytic, in the sense that it has a power series expansion near every point (power series will be discussed in the next section), and for this reason we also use the term **analytic** as a synonym for holomorphic. Again, this is in contrast with the fact that there are indefinitely differentiable functions of one real variable that cannot be expanded in a power series. (See Exercise 23.)

EXAMPLE 1. The function $f(z) = z$ is holomorphic on any open set in $\mathbb{C}$, and $f'(z) = 1$. In fact, any polynomial

$$p(z) = a_0 + a_1z + \cdots + a_nz^n$$

is holomorphic in the entire complex plane and

$$p'(z) = a_1 + \cdots + na_nz^{n-1}.$$

This follows from Proposition 2.2 below.

EXAMPLE 2. The function $1/z$ is holomorphic on any open set in $\mathbb{C}$ that does not contain the origin, and $f'(z) = -1/z^2$.

EXAMPLE 3. The function $f(z) = \bar{z}$ is not holomorphic. Indeed, we have

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\bar{h}}{h}$$

which has no limit as $h \rightarrow 0$, as one can see by first taking h real and then h purely imaginary.

An important family of examples of holomorphic functions, which we discuss later, are the power series. They contain functions such as e^z , $\sin z$, or $\cos z$, and in fact power series play a crucial role in the theory of holomorphic functions, as we already mentioned in the last paragraph. Some other examples of holomorphic functions that will make their appearance in later chapters were given in the introduction to this book.

It is clear from (1) above that a function f is holomorphic at $z_0 \in \Omega$ if and only if there exists a complex number a such that

$$(2) \quad f(z_0 + h) - f(z_0) - ah = h\psi(h),$$

where ψ is a function defined for all small h and $\lim_{h \rightarrow 0} \psi(h) = 0$. Of course, we have $a = f'(z_0)$. From this formulation, it is clear that f is continuous wherever it is holomorphic. Arguing as in the case of one real variable, using formulation (2) in the case of the chain rule (for example), one proves easily the following desirable properties of holomorphic functions.

Proposition 2.2 *If f and g are holomorphic in Ω , then:*

- (i) $f + g$ is holomorphic in Ω and $(f + g)' = f' + g'$.
- (ii) fg is holomorphic in Ω and $(fg)' = f'g + fg'$.
- (iii) If $g(z_0) \neq 0$, then f/g is holomorphic at z_0 and

$$(f/g)' = \frac{f'g - fg'}{g^2}.$$

Moreover, if $f : \Omega \rightarrow U$ and $g : U \rightarrow \mathbb{C}$ are holomorphic, the chain rule holds

$$(g \circ f)'(z) = g'(f(z))f'(z) \quad \text{for all } z \in \Omega.$$

Complex-valued functions as mappings

We now clarify the relationship between the complex and real derivatives. In fact, the third example above should convince the reader that the notion of complex differentiability differs significantly from the usual notion of real differentiability of a function of two real variables. Indeed, in terms of real variables, the function $f(z) = \bar{z}$ corresponds to the map $F : (x, y) \mapsto (x, -y)$, which is differentiable in the real sense. Its derivative at a point is the linear map given by its Jacobian, the 2×2 matrix of partial derivatives of the coordinate functions. In fact, F is linear and

is therefore equal to its derivative at every point. This implies that F is actually indefinitely differentiable. In particular the existence of the real derivative need not guarantee that f is holomorphic.

This example leads us to associate more generally to each complex-valued function $f = u + iv$ the mapping $F(x, y) = (u(x, y), v(x, y))$ from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Recall that a function $F(x, y) = (u(x, y), v(x, y))$ is said to be differentiable at a point $P_0 = (x_0, y_0)$ if there exists a linear transformation $J : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that

$$(3) \quad \frac{|F(P_0 + H) - F(P_0) - J(H)|}{|H|} \rightarrow 0 \quad \text{as } |H| \rightarrow 0, H \in \mathbb{R}^2.$$

Equivalently, we can write

$$F(P_0 + H) - F(P_0) = J(H) + |H|\Psi(H),$$

with $|\Psi(H)| \rightarrow 0$ as $|H| \rightarrow 0$. The linear transformation J is unique and is called the derivative of F at P_0 . If F is differentiable, the partial derivatives of u and v exist, and the linear transformation J is described in the standard basis of $\mathbb{R}^2$ by the Jacobian matrix of F

$$J = J_F(x, y) = \begin{pmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{pmatrix}.$$

In the case of complex differentiation the derivative is a complex number $f'(z_0)$, while in the case of real derivatives, it is a matrix. There is, however, a connection between these two notions, which is given in terms of special relations that are satisfied by the entries of the Jacobian matrix, that is, the partials of u and v . To find these relations, consider the limit in (1) when h is first real, say $h = h_1 + ih_2$ with $h_2 = 0$. Then, if we write $z = x + iy$, $z_0 = x_0 + iy_0$, and $f(z) = f(x, y)$, we find that

$$\begin{aligned} f'(z_0) &= \lim_{h_1 \rightarrow 0} \frac{f(x_0 + h_1, y_0) - f(x_0, y_0)}{h_1} \\ &= \frac{\partial f}{\partial x}(z_0), \end{aligned}$$

where $\partial/\partial x$ denotes the usual partial derivative in the x variable. (We fix y_0 and think of f as a complex-valued function of the single real variable x .) Now taking h purely imaginary, say $h = ih_2$, a similar argument yields

$$\begin{aligned} f'(z_0) &= \lim_{h_2 \rightarrow 0} \frac{f(x_0, y_0 + h_2) - f(x_0, y_0)}{ih_2} \\ &= \frac{1}{i} \frac{\partial f}{\partial y}(z_0), \end{aligned}$$

where $\partial/\partial y$ is partial differentiation in the y variable. Therefore, if f is holomorphic we have shown that

$$\frac{\partial f}{\partial x} = \frac{1}{i} \frac{\partial f}{\partial y}.$$

Writing $f = u + iv$, we find after separating real and imaginary parts and using $1/i = -i$, that the partials of u and v exist, and they satisfy the following non-trivial relations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

These are the **Cauchy-Riemann** equations, which link real and complex analysis.

We can clarify the situation further by defining two differential operators

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right).$$

Proposition 2.3 *If f is holomorphic at z_0 , then*

$$\frac{\partial f}{\partial \bar{z}}(z_0) = 0 \quad \text{and} \quad f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2 \frac{\partial u}{\partial z}(z_0).$$

Also, if we write $F(x, y) = f(z)$, then F is differentiable in the sense of real variables, and

$$\det J_F(x_0, y_0) = |f'(z_0)|^2.$$

Proof. Taking real and imaginary parts, it is easy to see that the Cauchy-Riemann equations are equivalent to $\partial f / \partial \bar{z} = 0$. Moreover, by our earlier observation

$$\begin{aligned} f'(z_0) &= \frac{1}{2} \left(\frac{\partial f}{\partial x}(z_0) + \frac{1}{i} \frac{\partial f}{\partial y}(z_0) \right) \\ &= \frac{\partial f}{\partial z}(z_0), \end{aligned}$$

and the Cauchy-Riemann equations give $\partial f/\partial z = 2\partial u/\partial z$. To prove that F is differentiable it suffices to observe that if $H = (h_1, h_2)$ and $h = h_1 + ih_2$, then the Cauchy-Riemann equations imply

$$J_F(x_0, y_0)(H) = \left(\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \right) (h_1 + ih_2) = f'(z_0)h,$$

where we have identified a complex number with the pair of real and imaginary parts. After a final application of the Cauchy-Riemann equations, the above results imply that

(4)

$$\det J_F(x_0, y_0) = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left| 2 \frac{\partial u}{\partial z} \right|^2 = |f'(z_0)|^2.$$

So far, we have assumed that f is holomorphic and deduced relations satisfied by its real and imaginary parts. The next theorem contains an important converse, which completes the circle of ideas presented here.

Theorem 2.4 *Suppose $f = u + iv$ is a complex-valued function defined on an open set Ω . If u and v are continuously differentiable and satisfy the Cauchy-Riemann equations on Ω , then f is holomorphic on Ω and $f'(z) = \partial f/\partial z$.*

Proof. Write

$$u(x + h_1, y + h_2) - u(x, y) = \frac{\partial u}{\partial x} h_1 + \frac{\partial u}{\partial y} h_2 + |h| \psi_1(h)$$

and

$$v(x + h_1, y + h_2) - v(x, y) = \frac{\partial v}{\partial x} h_1 + \frac{\partial v}{\partial y} h_2 + |h| \psi_2(h),$$

where $\psi_j(h) \rightarrow 0$ (for $j = 1, 2$) as $|h|$ tends to 0, and $h = h_1 + ih_2$. Using the Cauchy-Riemann equations we find that

$$f(z + h) - f(z) = \left(\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \right) (h_1 + ih_2) + |h| \psi(h),$$

where $\psi(h) = \psi_1(h) + \psi_2(h) \rightarrow 0$, as $|h| \rightarrow 0$. Therefore f is holomorphic and

$$f'(z) = 2 \frac{\partial u}{\partial z} = \frac{\partial f}{\partial z}.$$

2.3 Power series

The prime example of a power series is the complex **exponential** function, which is defined for $z \in \mathbb{C}$ by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

When z is real, this definition coincides with the usual exponential function, and in fact, the series above converges absolutely for every $z \in \mathbb{C}$. To see this, note that

$$\left| \frac{z^n}{n!} \right| = \frac{|z|^n}{n!},$$

so $|e^z|$ can be compared to the series $\sum |z|^n/n! = e^{|z|} < \infty$. In fact, this estimate shows that the series defining e^z is uniformly convergent in every disc in $\mathbb{C}$.

In this section we will prove that e^z is holomorphic in all of $\mathbb{C}$ (it is entire), and that its derivative can be found by differentiating the series term by term. Hence

$$(e^z)' = \sum_{n=0}^{\infty} n \frac{z^{n-1}}{n!} = \sum_{m=0}^{\infty} \frac{z^m}{m!} = e^z,$$

and therefore e^z is its own derivative.

In contrast, the geometric series

$$\sum_{n=0}^{\infty} z^n$$

converges absolutely only in the disc $|z| < 1$, and its sum there is the function $1/(1-z)$, which is holomorphic in the open set $\mathbb{C} - \{1\}$. This identity is proved exactly as when z is real: we first observe

$$\sum_{n=0}^N z^n = \frac{1 - z^{N+1}}{1 - z},$$

and then note that if $|z| < 1$ we must have $\lim_{N \rightarrow \infty} z^{N+1} = 0$.

In general, a **power series** is an expansion of the form

$$(5) \quad \sum_{n=0}^{\infty} a_n z^n,$$